

CANDIDATE
NAME

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INDEX NUMBER

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BIOLOGY 9648/02

Paper 2 Core Paper

15 September 2014
Monday

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

All working for numerical answers must be shown.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Calculators may be used

For Examiner's Use	
Section A	80
1	
2	
3	
4	
5	
6	
7	
Section B	20
8 / 9	
Total	100

Section A

Answer **all** the questions in this section.

- 1 Rubisco is an important enzyme responsible for carbon fixation in the Calvin cycle. As Rubisco has an optimum pH of 9, it is most active during the daytime when the pH in the stroma is high.

(a) Explain why Rubisco is most active during the daytime.

- In the daytime, presence of **light** cause **photophosphorylation** to occur
- energy released from ETC is used to **pump H^+ / actively transport H^+ from stroma into thylakoid space**
- (low H^+ concentration in the stroma) $\rightarrow$ reference to **active site**: the contact residues and/or catalytic residues in the active site are of the **correct charge** to bind the substrate molecules and catalyse the reaction respectively

OR **Ionic bonds** between R-groups are stabilized $\rightarrow$ shape of active site (most complementary to substrate)

[3]

In an investigation on limiting factors, activity of Rubisco was measured in isolated chloroplasts illuminated under optimum light intensity at different temperatures in the presence of 1mM and 30mM of sodium hydrogencarbonate solution ($NaHCO_3$). Fig. 1.1 shows the results of the investigation.

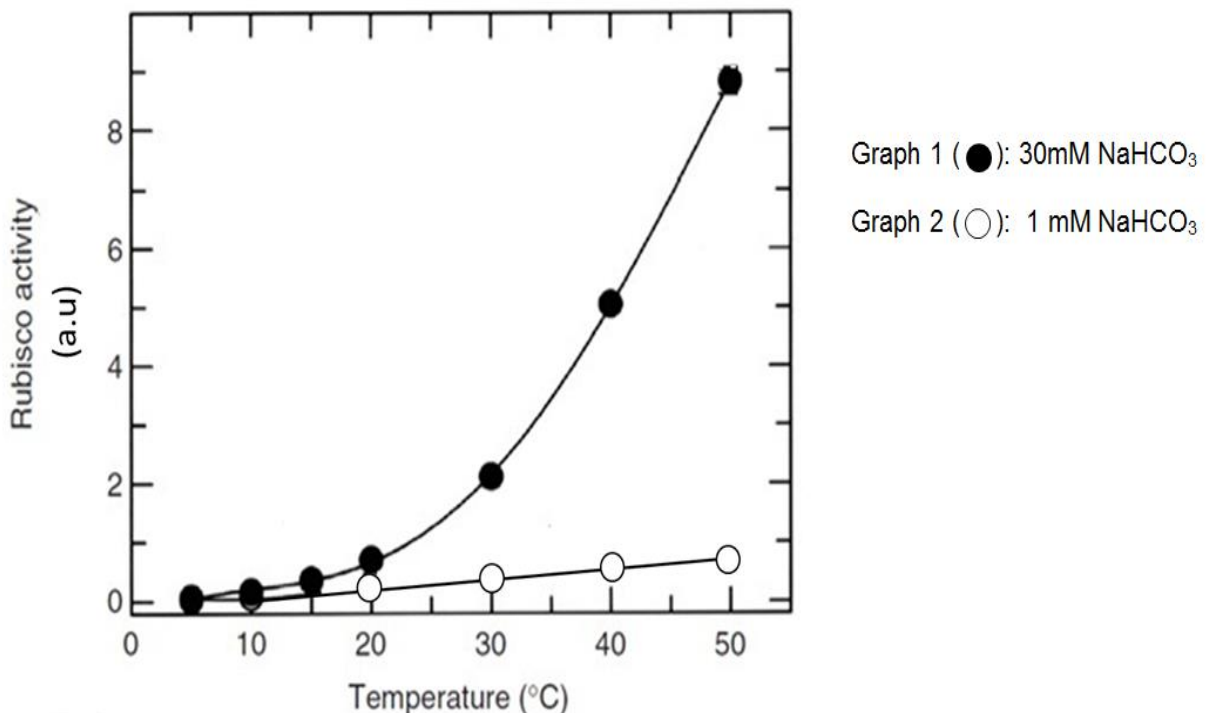


Fig. 1.1

(b) Explain the main factor limiting Rubisco activity in each graph.

Graph 1

- Limiting factor: Temperature
- When temperature **increases** from 20°C to 50°C, sharp/large **increase** in rubisco activity from 1 a.u to 9 a.u / accept other reference points
- As temperature increase, there is increasing **kinetic energy** in enzyme and substrate increasing **effective collisions** / increase **enzyme-substrate complex** formed per unit time

Graph 2

- Limiting factor: carbon dioxide concentration
 - Rubisco activity is higher in the presence of 30mM NaHCO₃ than 1mM NaHCO₃ at all temperatures / from temperatures 10°C to 50°C
 - hence increasing carbon dioxide increases **effective collisions** between CO₂, RuBP and rubisco
- increase **enzyme-substrate complex** formed per unit time [6]

- (c) In plants, the carbohydrates manufactured in photosynthesis are used to form macromolecule **A** as shown in Fig.1.2 and cellulose.

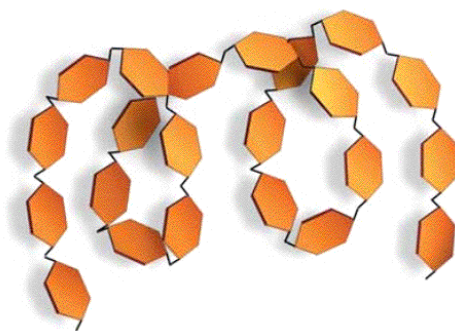


Fig. 1.2

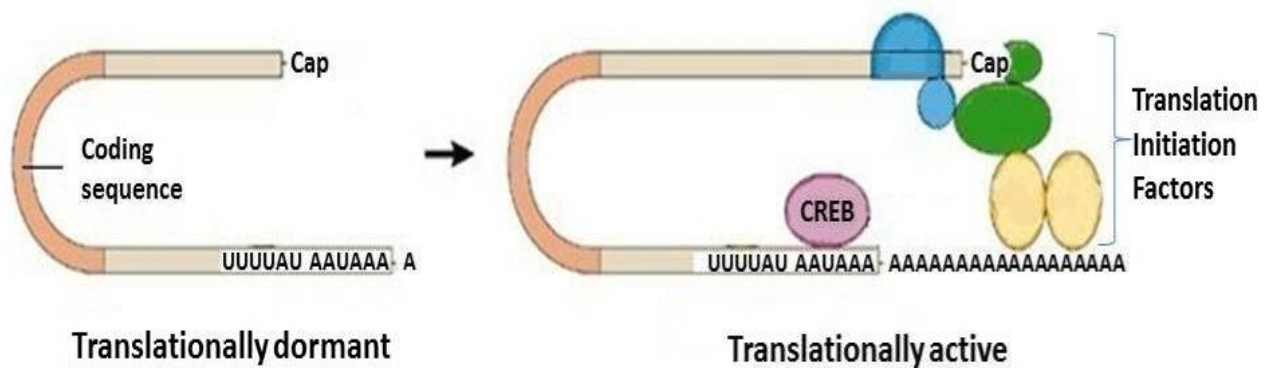
Describe how the structure of macromolecule **A** differs from the structure of cellulose.

- Structure A (Cellulose chain) is a **straight/linear** chain while structure B (amylose) is **coiled into helices**
- **Hydroxyl groups** projects in all directions from structure A while hydroxyl groups do not project in all directions from structure B
- Each successive glucose residue/ monomer in the structure A is rotated at 180° while glucose residue/monomer in structure B is not rotated/ at same orientation

[2]

[Total: 11]

- 2 Gene expression can be controlled at the translational level. Fig. 2.1 shows how the end of a mature mRNA in a frog zygote can be modified in the cytoplasm immediately upon fertilisation.



**diagram not drawn to scale*

Legend: CREB – A DNA binding protein that helps to elongate the ends of mature mRNA.

Fig. 2.1

- (a) With reference to Fig. 2.1, explain why such modification of mature mRNA is necessary for its stability and translation initiation.

Describe modification

- Poly A tails lengthening/Additional of adenine nucleotides by CREB to the 3' end (of mRNA). (reject: addition of adenine bases)

Accept idea that CREB recruits enzymes involved in poly-A tail lengthening.

Effects on stability

- longer time needed for poly-A tail to be shortened
- before mRNA degradation occur/to reaches critical length/increasing the half life of mRNA

misconception: many students wrong Poly (A) tailing “prevents” mRNA degradation.

Effects on translation initiation

- idea of longer tail Helps in the assembly/recruiting of the translation initiation factors at the 5' cap
- Reject: More time for assembly of translation initiation factors at the 5' cap.

- (b) Upon fertilisation, activation of amino acid also occurs immediately. Fig. 2.2 shows a molecule of tRNA and the enzyme that attaches the correct amino acid to it.

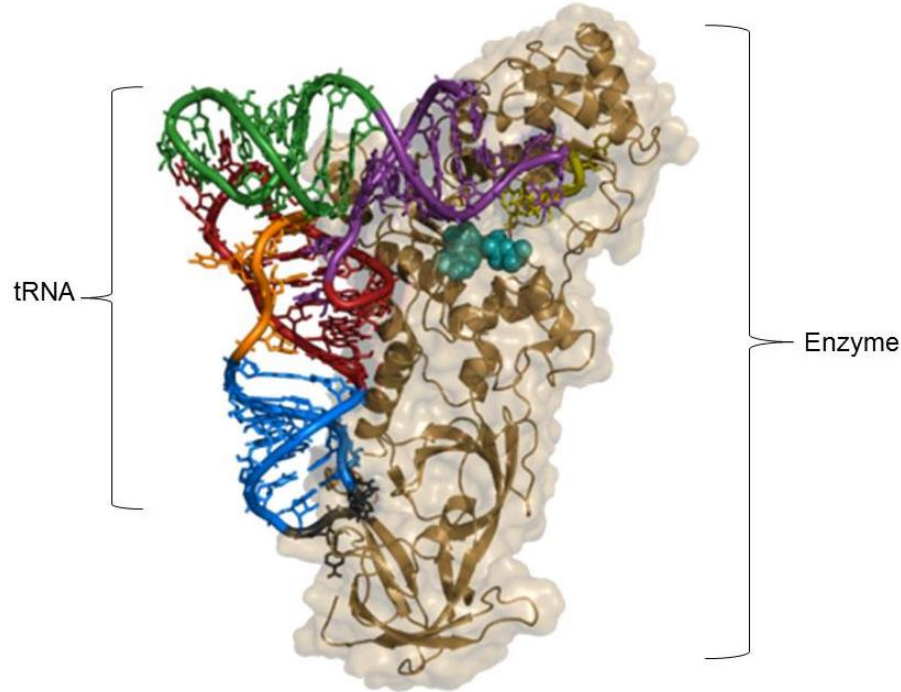


Fig. 2.2

- (i) Describe the role of tRNA in translation.

- c) anti-codon undergo complementary base pairing with codon on mRNA.
- d)
- e) bring specific amino acids (idea of a certain idea) to be added to the growing polypeptide chain/protein + on the ribosome

[2]

- (ii) It was found that an amino acid substitution in the enzyme halts zygote development.

With reference to Fig. 2.2, explain why an amino acid substitution can affect the function of the enzyme.

- f) Amino acid substitution leads to a different R group with a different R grp properties/size/charges → leading to a change in R grp interactions (→ change in 3D shape of the enzyme)
- g) Shape of the active site no longer complementary to the (anti-codon) of the tRNA and amino acid. No covalent/ester bond formed between the amino acid and tRNA

[3]

- (iii) Describe in what ways the helical structures that are shown in Fig. 2.2 in both tRNA and enzyme differ.

tRNA	Enzyme
Forms double helix (from a single strand)	Forms alpha helix
Maintained by Hydrogen bonds form between complementary bases (Rejected bonds formed between nucleotides → we are looking at bonds formation between specific components of a monomer)	Hydrogen bonds form between CO of one amino acid and NH group of another amino acid (misconception – alpha helix is maintained by the four different bonds !)
Irregular helical structure due to RNA folding back on itself to form (hydrogen) bonds between different part of the <u>same</u> strand	Regular helical structure due to (hydrogen) bonds formed between one amino acid and the next amino acid that is four residues ahead

Ref to N2006 P2 Q3.

Many candidates explained how the “structures” differed rather than how the helices differed.

Reject comparisons that focuses on monomers, types of bonds between monomers, and size , level of folding

Reject double stranded vs single stranded

[3]

- (c) Edeine is an antibiotic that inhibits protein synthesis. In an investigation, edeine is added to a cell extract obtained from a developing frog embryo. It was found that edeine stops protein synthesis after a short lag. Analysis of the edeine-inhibited cell extract showed that no polyribosomes remained at the time protein synthesis had stopped. Instead, all the mRNA was found associated with small ribosomal subunit and initiator tRNA.

Suggest how edeine inhibits protein synthesis.

- Preventing the large ribosomal subunit from binding to the small ribosomal subunit/mRNA/initiator tRNA /from ribosome assembly
- Reject: Edeine degrades large ribosomal subunit → antibiotic is protein molecule. It cannot directly degrades ribosome.

Reject: prevent the assembly ribosomal subunits to form ribosome.

(No translation initiation complex formed/inhibit initiation of protein synthesis)

[1]

[Total: 12]

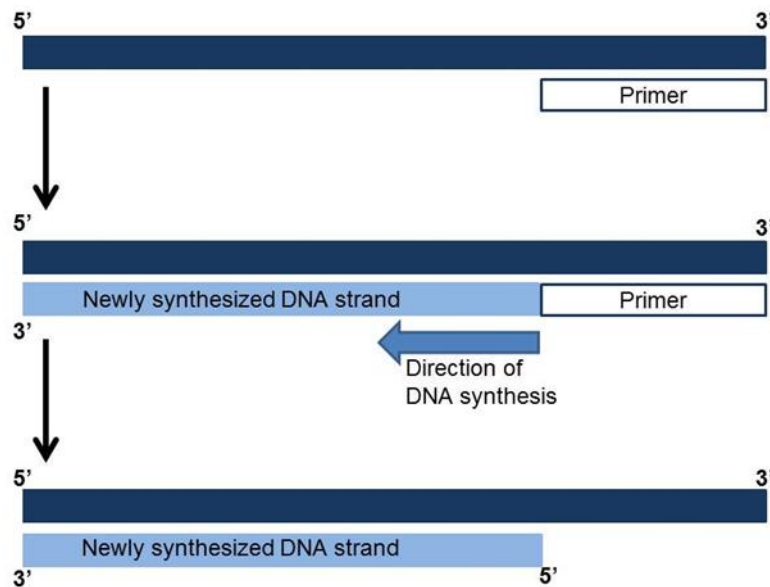


Fig. 3.1

(a) Explain why the end-replication problem does not occur in bacteria.

- Bacterial chromosomes are circular DNA molecules;
- Lagging strands can be completely replicated / Presence of free 3'OH group at the end of a pre-existing strand for DNA polymerase to add DNA nucleotides to;

[2]

(b) With reference to Fig. 3.1, explain how the end-replication problem may lead to fusion of chromosome ends.

- Ref. to the 3' overhang produced after DNA replication / part of the DNA strand at the 3' end is unpaired / single-stranded;
- complementary base pair with the 3' overhang / unpaired strand / single-stranded DNA at the end of another chromosome;

[2]

(c) State **two** differences between transcription and the process of lengthening of telomeres.

- RNA is used as a template to lengthen the telomere while DNA is used as a template strand in transcription;
- The product synthesised is DNA in the lengthening of the telomere while the product synthesised is RNA in transcription;
- Enzyme involved is telomerase in lengthening of telomere while enzyme involved in transcription is RNA polymerase
- Monomers used in lengthening of the telomere are DNA nucleotides while monomers used in transcription are RNA nucleotides;

[max 2m]

[2]

Fig. 3.2 shows the results of an experiment in which two yeast colonies with different ADE2

gene loci are cultured.

Each of these colonies contains about 100,000 cells produced from a single yeast cell through multiple rounds of cell division. The cells closer to the centre of a colony are of earlier generations while those on the edge of a colony are of later generations.

Expression of ADE2 gene prevents accumulation of red pigments and results in cells being white in colour. When the ADE2 gene is at its normal chromosomal locus, its expression causes the colony colour to be white (Colony 1). When a segment of the chromosome is inverted such that the ADE2 gene locus is near the telomere, the gene is not expressed and the colony has a red colour (Colony 2).

Some proteins bind to telomeres. This causes telomere and adjacent DNA to wind around these proteins, causing chromosomal ends to be in a condensed state. The ADE2 gene is therefore silenced.

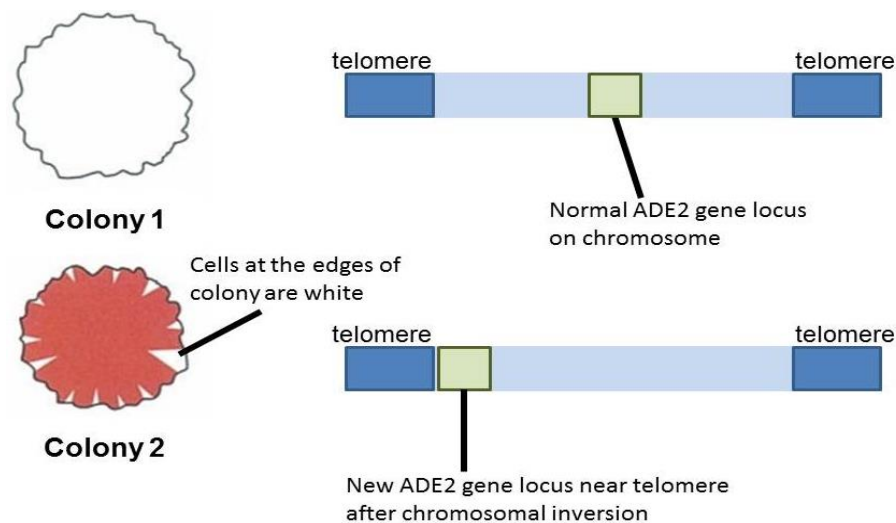


Fig. 3.2

As Colony 2 grows, it is observed that some cells of the later generations at the edges of the colony appear white in colour.

(d) Explain why cells at the edges of Colony 2 appear white in colour.

- Telomere length shortens as cells undergo repeated rounds of DNA replication;
- Cells at the edge of colony have short telomeres such that the DNA is unable to bind to / coil / wind around proteins;
- RNA polymerase and transcription factors are able to bind to promoter sequence to form transcription initiation complex;
- Transcription and subsequently translation of ADE2 gene can occur again, leading to white cells;

[4]

Colony 2 stops growing after a certain period of time as the yeast cells stop dividing. When the yeast cells are artificially induced to continue dividing, the new cells at the edges of the colony appear red in colour.

(e) Explain why the yeast cells stop dividing and yeast cells produced from subsequent cell division appear red in colour.

- Telomeres shortened to critical length;
- triggering cell senescence; /limit no. of cell division.
- for subsequent cell division, ADE2 gene is being eroded and no longer expressed;

[3]

[Total: 13]

4 Fig. 4.1 shows the reproductive cycle of the Human T-lymphotropic virus type I (HTLV-I), which can integrate its viral nucleic acid into the host cell's genome. This virus has been implicated in several kinds of diseases including very aggressive adult T-cell lymphoma, the cancer of T-lymphocytes.

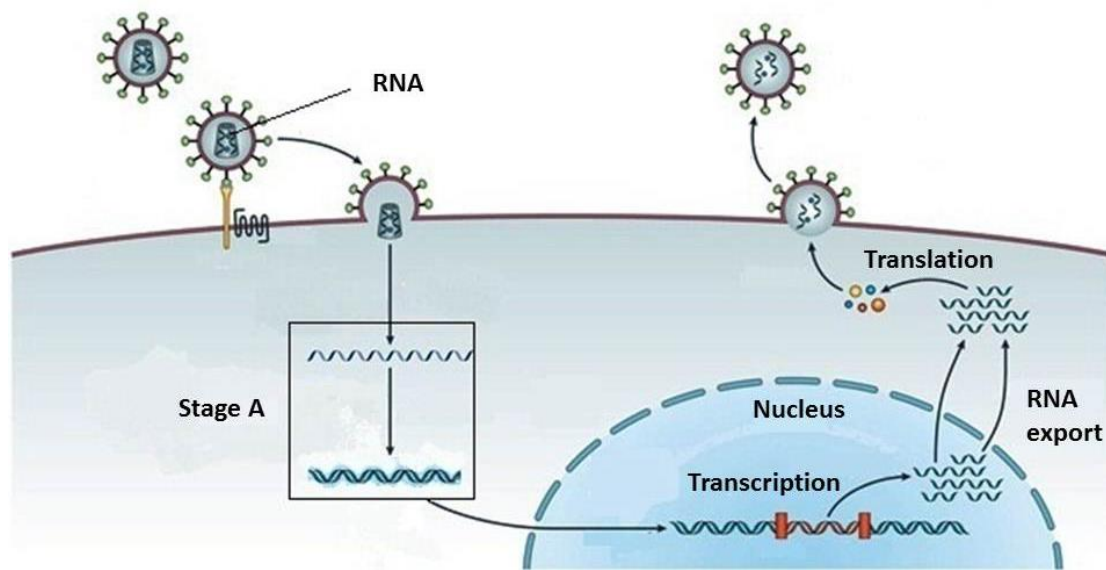


Fig. 4.1

(a) Contrast the mode of entry of a lytic phage with HTLV-I.

Lytic phage	HTLV-I
<u>Tail fibre</u> of T4 binds to <u>complementary specific receptors</u> on <u>bacterial cell membrane</u>. (R: Base plate pin)	<u>Glycoprotein</u> binds to <u>specific receptor</u> on host cell membrane. (R; gp120) /CD4 receptors Accept when there is mention of complementary receptors at least once. Some students wrongly wrote that the the glycoproteins of HTLV I is haemagglutinin, which bind to sialic acid receptor.
Contraction of tail sheath/ The tail core protein is driven through the wall and the membrane + phage DNA injected.	<u>Fusion</u> of the viral's envelope (reject: membrane) with the host's cell membrane and capsid enters. Many thought that the process of entering

[2]

	the cell is receptor mediated endocytosis .
--	---

A handful of students still did not give point to point differences.

Reject: phage capsid left outside the cell for lytic phage vs nucleocapsid enters the cell.

Some don't know the meaning of contrast, which means differences

Mode of entry → differences in the process

Eg: genome is injected directly into the host cells vs genome is released via fusion of membrane.

Nucleocapsid left outside + only genome enters vs nucleocapsid enters the cell.

No fusion vs fusion of membrane

Degradation of membrane vs no degradation of membrane.

Must show a diagram of endocytosis

(b) With reference to Fig. 4.1,

(i) Explain the significance of stage **A** to HTLV-I.

- Virus uses reverse transcriptase to transcribe the RNA template into a complementary single stranded DNA. (pls spell out ss and ds)
- Reverse transcriptase (Not: DNA polymerase) then synthesises a complementary DNA strand, forming a double-stranded DNA./allow for complementary base pairing to form a double-stranded DNA.
Mention template and complementary strand at least once.
- So can be integrated into the double stranded DNA.

Many talks abt the dividing of the cells.

Use of the viral genome for transcription and translation

[3]

(ii) Explain why human glycoproteins can be found on the HTLV-I's envelope.

- (reject Nucleocapsid buds off) virus buds off (Reject: exocytosis) from the host cell membrane
- acquiring their envelope from portions of the host's cell membrane with embedded human glycoproteins. (Idea of envelop is obtained from the host cell membrane during budding and glycoproteins are already embedded on the host cell membrane)

Reject vague phrasing like, "since the envelope is obtained from human host cells, it contains human glycoproteins)

Reject: During exocytosis , budding of virus occurs.

[2]

Note the differences between invagination and evagination.

- (c) Explain how the integration of the HTLV-I's nucleic acid into the host's genome aids in the reproduction of virus while causing cancer to the host cell.
- Integration (of viral DNA into the host chromosome) disrupts the proto-oncogene, converting it to oncogene.
 -
 - Hyperactive proteins produced leading to excessive cell division/ excess amount of normal proteins to be produced leading to excessive cell division (if promoter/enhancer affected)
 -

or

- integration disrupts a tumour suppressor gene.
- no longer encode for the protein to inhibit cell division/ for apoptosis → uncontrolled cell division.

and

- viral DNA can replicate along/ together with the host cell DNA as host cell divides
- Idea of results in production of more cells that carries the viral DNA without killing the host cell. (reject: virus propagate)

Reject: transcription and translation of the viral genome (not significant)

Some say the HTLV I virus undergo lytic cycle. This note that lytic cycle is reserved for lytic bacteriophage.

[4]

[Total: 11]

- 5 Pure-breeding fruit flies, *Drosophila melanogaster*, with wild-type bodies and vestigial wings were crossed with pure-breeding flies with black bodies and wild-type wings.

The F_1 all had wild-type bodies and wild-type wings.

Female flies from the F_1 were crossed with black bodied, vestigial-winged males. 2876 offspring flies were produced.

Fig. 5.1 shows flies with the phenotypes described.

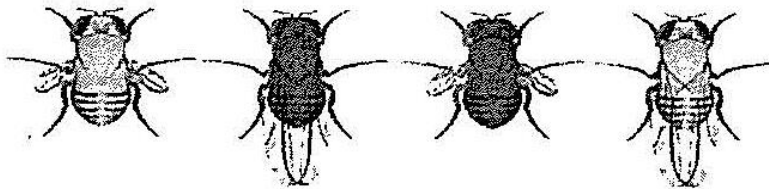


Fig. 5.1

The genes coding for body colour and wing length are thought to be independently segregating.

- (a) State the expected number of F_2 flies with each phenotype in this dihybrid cross.

719 flies for each phenotype;

[1]

(b) The observed numbers of F_2 flies with each phenotype were as follows:

wild-type body, vestigial wings	1277
wild-type body, wild-type wings	143
black body, vestigial wings	152
black body, wild type wings	1304

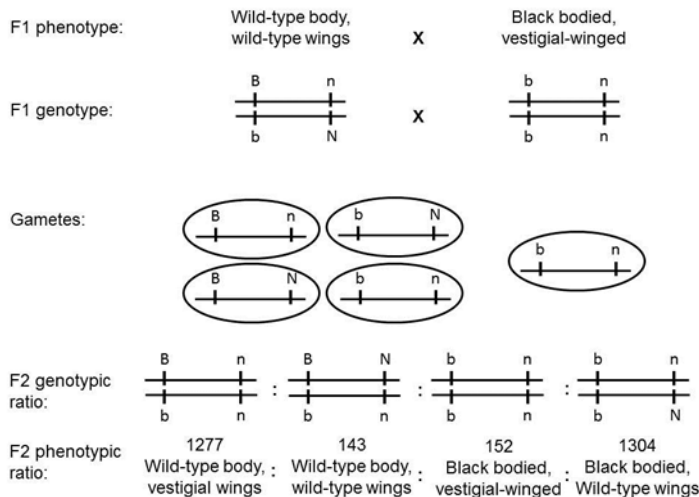
Draw a genetic diagram to explain the observed results of the cross between F_1 female flies and black bodied, vestigial-winged male flies.

Use the following symbols,

B/b for gene locus controlling body colour; N/n for gene locus controlling wing length.

Definition

- Let B represent allele coding for wild-type body;
- Let b represent allele coding for black body;
- Let N represent allele coding for normal wing;
- Let n represent allele coding for vestigial wing;



- Definition of alleles;
- Correct genotypes of parents;
- Correct gametes (must be circled);
- Correct linking of F_2 genotype to phenotype;

[4]

(c) (i) Outline how the significance of the difference between observed and expected results can be tested (Formulae not required).

- Carry out chi-squared test;
- Determine probability of difference occurring due to chance alone;

Accept if students mention that the calculated chi-square value is lower than the critical chi-square value at $p = 0.05$, the difference is not significant or due to chance

[2]

alone /converse agreement.

(ii) Explain why there is a difference between the observed results and the expected numbers you have provided in (a).

- two gene loci are linked / on the same chromosome;
- gene loci located far enough from each for **crossing over** / **chiasmata formation** to occur;
- larger parental types and fewer recombinants due to crossing over / chiasma formation being a **chance event** that has **low probability of occurring**; [3]

(d) Explain how you could determine that the genotype of a F_2 fly with wild type body and wild type wings is heterozygous at both gene loci.

- Cross the fly with wild-type body and wild-type wings with a fly with black body and vestigial wings;
- If there is presence of offspring flies which have black bodies and vestigial wings, then the F_2 fly is heterozygous at both gene loci; [2]

[Total: 12]

6 The tiger, *Panthera tigris*, is the largest and most distinctive cat in the world.

Tigers are further classified into a number of sub-species (races) based on marked phenotypic differences, such as body size and colour. Fig. 6.1 shows the distribution of the sub-species 100 years ago and in 2004. The names of the sub-species are shown on the map.

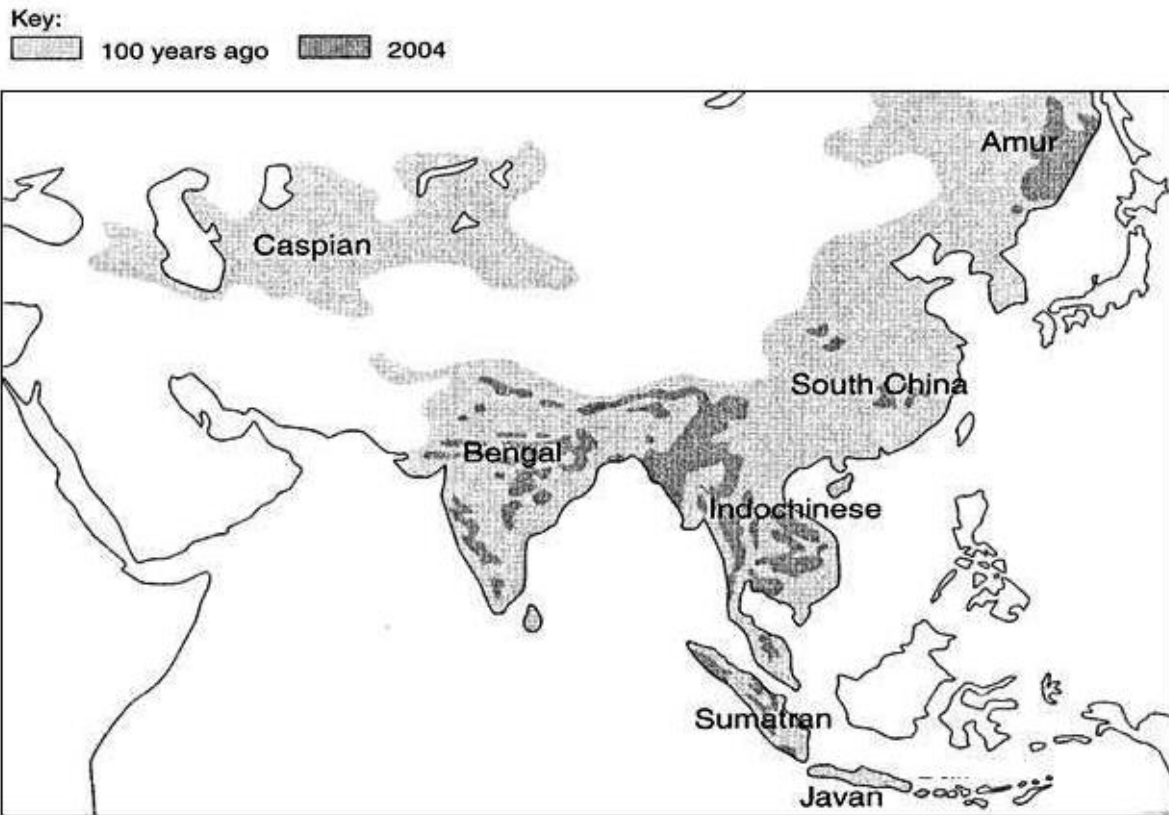


Fig. 6.1

(a) Describe the changes shown in Fig. 6.1.

- Fewer sub-species in 2004 than 100 years ago, decrease from 7 to 5 sub-species/ accept listing of species present 100 years ago and then in 2004
- Habitats/ population range spanned/covered a larger area/more widely distributed across a wider geographical area 100 years ago as compared to in 2004;
- Populations more fragmented/sparsely distributed/ more scattered in 2004 compared to 100 years ago;
- Caspian and Javan tiger sub-species disappeared/ went extinct in 2004;
- Largest reduction of South China sub-species by 2004;

[max 2]

•

[2]

(b) Explain how the distinct phenotypic differences between the sub-species may have arisen.

- Mutations in gene sequences give rise to new alleles which code for new phenotypes/ traits OR genetic variation gives rise to phenotypic variation amongst ancestral tiger population (Reject: if student did not mention in which population variation exists);
- (Different geographical locations/ecological niches), exert different selection pressure (Reject: selection factors);
- Different traits/ phenotypes (Reject: features, alleles) have selective advantage (at different locations);
- Survive, reproduce and pass on alleles coding for advantageous traits to subsequent generations/ offspring (Reject: advantageous alleles);
- Alleles coding for selectively advantageous traits increase in (allele) frequency/ Change in allele frequency/ gene pool diverge/ genetic difference between gene pools increase;

- Populations **geographically isolated** (*Reject: geographically separated*), therefore little/limited/prevent/no gene flow between different populations;
- **In small populations**, **genetic drift operate** (to **alter/change allele frequency**) and result in significant differences in gene pool;
- AVP (Founder's effect);

[max 4]

- (c) Suggest **two** reasons why these populations of tigers are classified as different sub-species rather than different species.

- Can interbreed to produce fertile, viable offspring (although geographically isolated);
- DNA sequence of many genes very similar;
- Morphological/physiological/behavioural/biochemical features (name 2 out of the 4) are very similar (but not different enough to be classified as distinct species);
- Occupy same ecological niche;

[max 2]

- (d) Using the biogeographical evidence in Fig. 6.1, deduce the evolutionary relationship between the Indochinese, South China and Javan sub-species.

- Indochinese and South China subspecies are more closely related with each other than with the Javan subspecies;
- Indochinese and South China sub-species found on same land mass/ piece of land, but Javan sub-species are separated from the two/ OR Indochinese and South China sub-species are located more closely/closer together than Javan sub-species/ OR Similar/ closely related species are found in habitats located close together;

[2]

- (e) Differences in the mitochondrial DNA sequences between species belonging to the genus *Panthera* were measured, and it was found that the snow leopard is most closely related to the tiger.

Explain how a molecular clock can be used to derive this evolutionary relationship.

- Molecular clock is based on the assumption that selectively neutral mutations occur at a (relatively) constant rate;
- Molecular clock for gene on mitochondrial DNA can be calibrated by plotting number of nucleotide differences against the dates of evolutionary branch points which are known from fossil records
- Sequence of a particular common/ubiquitous gene (or give a named example such as cytochrome C gene) on mitochondrial DNA is compared between the snow leopard and tiger;
- The smaller the number of differences in mitochondrial DNA sequence for the particular common gene, the shorter the time since divergence of tiger and snow leopard from common ancestor;

[3]

[Total:13]

- 7 Fig. 7.1 is an electron micrograph showing the synapse at an axodendritic junction between a pre-synaptic neurone and a post-synaptic neurone.

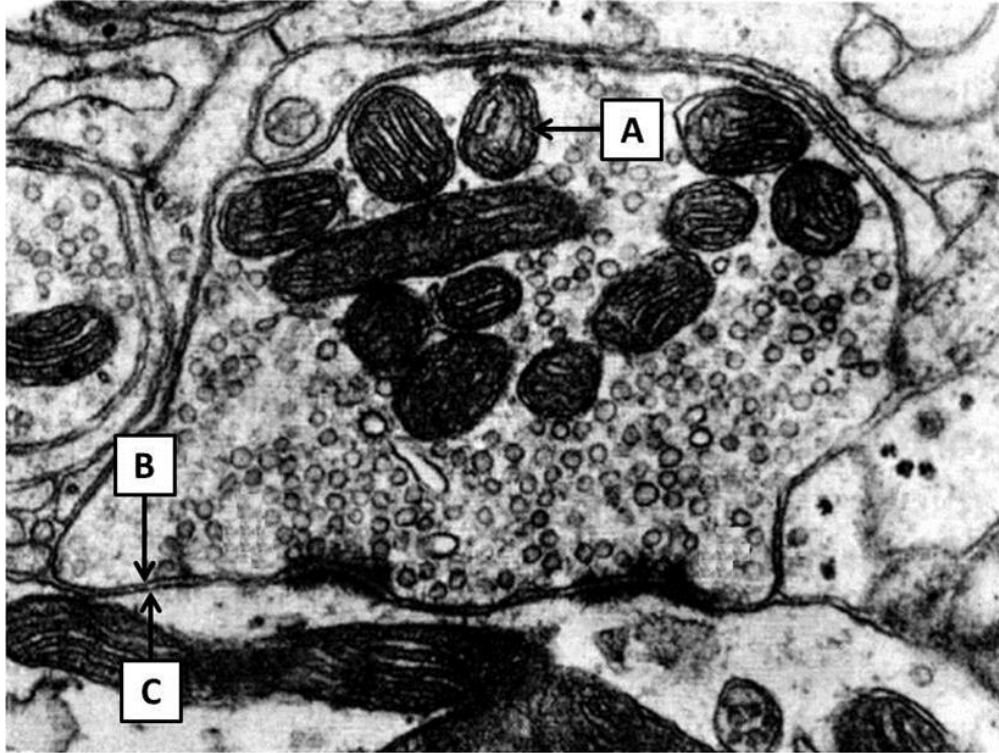


Fig. 7.1

- (a) Explain why a large number of organelle A is present in the synaptic knob.

- Organelle A is the mitochondrion (must always name unknowns in answers) → required to produce large amount of ATP through aerobic respiration / oxidative phosphorylation;

[1m]

- ATP/energy required for **active transport** of Na^+ and K^+ ions by **sodium potassium pumps**;
- ATP/energy required for **active transport** of **calcium ions** out of the cell;
- ATP/energy required for **transport/migration** of **synaptic vesicles** to **fuse** with the **synaptic membrane** to release neurotransmitters;
- ATP/energy required for exocytosis of neurotransmitters;

[max 2m]

•

[3]

- (b) Explain how the differences between structures B and C allow nerve impulses to travel in only one direction across the synapse.

- Structure B is pre-synaptic membrane while C is post-synaptic membrane (*Reject: knob*);
- **Voltage-gated Ca^{2+} ion channels** only found on Structure B (pre-synaptic membrane) and not C (post-synaptic membrane);
- **Ca^{2+} influx/ diffusion** into **pre-synaptic knob** (*Reject: membrane*) **only** when voltage [5]

Ca^{2+} ion channels open during depolarisation, leading to **release of neurotransmitters / acetylcholine** via exocytosis from **pre-synaptic neurone**;

- **Ligand-gated Na^+ channel proteins** are found only on Structure C and not B;
- neurotransmitter can only move from presynaptic membrane to **bind with receptors** on **post-synaptic membrane** to **trigger depolarisation / action potential** of post-synaptic membrane;

[Total: 8]

Section B

Answer **EITHER 8 or 9.**

Write your answers on the separate answer paper provided.

Your answer should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in section (a), (b) etc., as indicated in the question.

- 8 (a)** Describe the structure of phospholipids and explain how it enables them to form the cell surface membrane. [4]

Structure:

- It is made up of one glycerol, two fatty acids and one phosphate group (PO_4^{3-}).
- The fatty acids are joined to the glycerol group via ester bonds, while the phosphate group is joined to glycerol via phosphoester bond;
- Since it has both hydrophobic and hydrophilic regions, it is also known as an amphipathic molecule.

[max 2m]

Formation of cell surface membrane:

- Hydrophilic phosphate heads interact with the aqueous medium/environment
- Hydrophobic interactions between fatty acid chains/tails
- Hence, in an aqueous environment, phospholipids self-assemble into a bilayer, forming a hydrophobic boundary/barrier that separates the living cells from its surrounding.

[max 2m]

- (b)** Explain how fluidity of biological membranes can be maintained and the importance of fluidity to membrane function. [8]

How fluidity is maintained:

- Definition of fluidity: phospholipids and proteins free to move within layer / transversely;
- Membrane fluidity can be regulated by regulating the level of cholesterol and phospholipids with unsaturated hydrocarbon tails in the membrane;
- Membranes freeze at a lower temperature if it has a **higher proportion of phospholipids with unsaturated hydrocarbon** tails and cholesterol; (Accept reverse argument)
- Unsaturated hydrocarbon tails have kinks that keep the phospholipid molecules in the membrane from packing close together at lower temperatures, enhancing membrane

fluidity; (Accept reverse argument)

- Cholesterol prevents close packing of phospholipids in the membrane, preventing membrane from freezing at lower temperatures;
- Membrane prevented from being overly fluid / integrity of the membrane is maintained at higher temperature as cholesterol restricts phospholipid movement through its interactions with phospholipids;
- Stabilising the lipid bilayer through hydrophobic interactions;

[max 4m]

Importance:

- ref. to need for invagination / pinching in of the membrane during cytokinesis / cell division;
- ref. to invagination of the membrane during phagocytosis / pinocytosis / endocytosis;
- ref. to formation of vesicles for the transport / trafficking of proteins from rough ER to Golgi apparatus / other membrane-bound organelles;
- ref. to formation of vesicles from *trans* face of the Golgi apparatus during protein sorting;
- ref. to fusion of vesicles from endoplasmic reticulum / containing proteins to form cis-face of the Golgi apparatus for protein modification;
- ref. to fusion of vesicles with cell surface membrane to release secretory molecules from the cell / exocytosis;
- ref. to fusion of endocytotic vesicles with lysosomes;
- ref. to membrane fluidity allowing movement of protein molecules embedded on the cell surface membrane for dimerisation of receptors;
- ref. membrane fluidity to allow locomotion of unicellular organisms e.g. amoeba;
- ref. diffusion of small or non-polar molecules through the membrane, e.g. water / oxygen / carbon dioxide (at least one named example);
- ref. to transmembrane transport proteins changing conformation, such as sodium-potassium pump / ligand-gated sodium ion channel / voltage-gated sodium ion channel (at least one named example);

[max 4m]

(c) Describe the roles of membranes in photosynthesis and in respiration.

- stalked particle / ATP synthase embedded in the inner membrane of mitochondrion and thylakoid membrane to allow harnessing of proton motive force to synthesise ATP from ADP and phosphate groups;
- proton gradient generated across the mitochondrion inner membrane and thylakoid membrane for ATP synthesis due to impermeability of membrane;
- holds electron carriers of the electron transport chain to transport electrons, releasing energy to pump H^+ into the mitochondrial inter-membrane space and thylakoid space
- holds proton pumps coupled to the electron carriers that actively transports H^+ against a concentration gradient;
- thylakoid membrane holds the photosynthetic pigments that absorb light energy during photosynthesis;
- outer and inner mitochondrial membrane encloses the inter-membrane space in mitochondrion and thylakoid membrane encloses the thylakoid space to allow a buildup of H^+ ; (same point as point 2)
- inner membrane of the chloroplast and both membranes of mitochondrion contains transport proteins to allow transport of metabolites e.g. triose phosphate;
- mitochondrion outer membrane contains enzymes for various metabolic reactions e.g.

tryptophan degradation;

- the envelopes of mitochondrion and chloroplast allows for compartmentalisation of the cell as the mitochondrion and chloroplast enzymes require different / specific conditions from the rest of the cell to function;

[max 8m]

[Total: 20]

Or

- 9 (a) Outline the ways in which transfer of DNA occur from one bacterium to another. [7]

Transformation

- Bacterial cells take up **foreign** DNA from the **surrounding** medium **via a cell surface receptor**
- Incorporating / integration** of foreign DNA into its own DNA (**via homologous recombination**)

Transduction - Award marks either for generalized or specialised transduction only

- A **phage infects** and **injects** its DNA into a bacterium and **replicate** in the cell.
- Due to mistakes in the phage's reproductive cycle, a **small piece of bacteria DNA was incorporated** into the **phage capsid**
- The resulting transducing phages **infect other bacteria** and newly infected cell acquires the **original bacterial DNA** / original bacterial DNA integrates into the DNA of the recipient

Conjugation

- Attachment of F+ and F- bacterium via sex pilus made by F+ cell
- Sex pilus retracts, the two bacteria** cells come into **physical contact** via **conjugation tube/mating bridge**
- Single strand of F plasmid breaks** at **origin of transfer** and move into recipient bacterium through conjugation tube

(any 7)

- (b) Compare prokaryotic control of gene expression with the eukaryotic model. [8]

Max 6

Level	Prokaryotic Control	Eukaryotic Control
Chromosomal level	No association with histones, hence no histone modification possible (for control of gene expression at chromosomal level)	Histone acetylation and methylation (accept modification) can occur, resulting in conversion between euchromatin and heterochromatin states, controlling access of transcription machinery to the DNA
Transcriptional level	Transcription initiation requires fewer regulatory proteins (e.g. catabolite activator protein, and repressor to bind to RNA polymerase)	Transcription initiation requires many different regulatory proteins e.g. basal TFs/ mediator proteins/ activators to form transcription initiation complex
	Regulatory sequence, operator , is <u>close</u> to the promoter	Regulatory sequence, control elements , can be <u>close</u> (proximal control elements) or <u>far away</u> (distal control elements) from promoter

	Multiple related genes of the same metabolic pathway are <u>regulated together</u> by a <u>single</u> promoter in an operon system (→ polycistronic mRNA)	<u>Each</u> individual gene is controlled by its <u>own promoter</u> (→ monocistronic mRNA)
Post-Transcriptional and Translational level	No post Transcriptional control and minimal translational control because transcription and translation occur simultaneously due to the absence of a nuclear membrane	Nuclear membrane separates the processes of transcription and translation, hence post-transcriptional and translational level of control possible
	No mRNA processing	mRNA processing must occur before mRNA can leave nucleus to cytoplasm for translation
	→ leads to lower stability of RNA transcript and its degradation within seconds / minutes → no alternative splicing	→ 5'cap and 3' poly-A tail increase transcript stability / prevent transcript degradation → alternative splicing possible to allow different proteins to be encoded by 1 gene
	Minimal control at translational initiation	Control at translational initiation can also occur via binding of translation repressor proteins at mRNA, regulation of mRNA poly-A tail in cytoplasm and activation/inactivation of translational initiation factor (any 1 mechanism)
Post-translational level	No/minimal post-translational modifications	Protein processing (such as cleavage and addition of chemical groups) determine functional abilities of proteins
	Degradation of proteins does not involve ubiquitin	Ubiquitinylation of proteins targets specific proteins for degradation by proteasome

Similarities (max 3)

- Most significant / main level of control at transcriptional level
- Rate of transcription control is affected by the affinity of the RNA polymerase to the promoter
- Formation of polyribosomes to speed up the rate of translation
- Degradation of mRNA / proteins as a method of terminating response

(c) Suggest with reasons, the evolutionary origin of mitochondrion and chloroplast.

[5]

Evolutionary Origins (max 3)

- ancestors of mitochondria were **oxygen-using non-photosynthetic prokaryotes/aerobic prokaryotes**.
- ancestors of chloroplasts were **photosynthetic prokaryotes**.

Idea of symbiosis

- An early ancestor of eukaryotic cells **engulfed** (one of these) prokaryote
- Eventually, the engulfed cell formed a symbiosis relationship with the host cell (in which

it was enclosed) becoming an **endosymbiont**

- Gives a selective advantage to prokaryote → synthesize ATP/Photosynthetic ability
- Over the course of evolution, the host cell and its endosymbiont merged into a single organism, a **eukaryotic cell** with a **mitochondrion** and/or **chloroplast**

Reasons (max 3)

- Mitochondria and chloroplasts are bounded by a **double membrane**
- the inner membrane is the original membrane of the once independent prokaryotes while the outer one is the food vacuole (phagosomal membrane/endocytic vesicle membrane) it was enclosed in initially

OR

- the double membrane enclosing chloroplasts and mitochondria are both derived from those the ancestral bacteria, and the phagosomal membrane disappeared during organelle evolution
- Like prokaryotes, mitochondria and chloroplasts contain **70S ribosomes** and **circular DNA (not enclosed in nucleus)** → *must mention both to gain mark*
- **Undergo transcription, translation and DNA replication**
- the mitochondria and chloroplasts are **autonomous organelles** that grow and reproduce within the cell (independent of whether they undergoes cell division or not)
- mode of reproduction of mitochondria and chloroplast is also via a process similar to **binary fission** in prokaryotes
-

[Total:20]